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Machine Learning Models for Cardiovascular Disease Events Prediction*

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***Abstract*—Cardiovascular diseases (CVDs) are among the most serious disorders leading to high mortality rates worldwide. CVDs can be diagnosed and prevented early by identifying risk biomarkers using statistical and machine learning (ML) models, In this work, we utilize clinical CVD risk factors and biochemical data using machine learning models such as Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), Naïve Bayes (NB), Extreme Grading Boosting (XGB) and Adaptive Boosting (AdaBoost) to predict death caused by CVD within ten years of follow-up. We used the cohort of the Ludwigshafen Risk and Cardiovascular Health (LURIC) study and 2943 patients were included in the analysis (484 annotated as dead due to CVD). We calculated the Accuracy (ACC), Precision, Recall, F1-Score, Specificity (SPE) and area under the receiver operating characteristic curve (AUC) of each model. The findings of the comparative analysis show that Logistic Regression has been proven to be the most reliable algorithm having accuracy 72.20 %. These results will be used in the TIMELY study to estimate the risk score and mortality of CVD in patients with 10-year risk.**

***Keywords*—Machine Learning Models, Cardiovascular Disease, Risk Scores, 10-Year Risk of CVD.**

I. INTRODUCTION

Cardiovascular disease is a hazardous disorder for humans, accounting for approximately 17.9 million deaths every year worldwide, which represents 32% of all global deaths [1]. CVD is expected to account for >23.6 million deaths annually by 2030 [2]. In Europe, there were 1.68 million deaths resulting from cardiovascular disease in 2016, which was equivalent to 37.1 % of all deaths [3]. Several risk factors can trigger cardiovascular disease, such as diabetes mellitus, LDL cholesterol, blood pressure, irregular pulse rate, physical activity, unhealthy diet, family history of CVD and ethnic background. Cardiovascular disease can be predicted using multiple tests. However, the lack of expertise of medical staff can make early diagnosis difficult [4]. It is necessary to evaluate the risk of cardiovascular disease for prevention, primary and secondary. There are several statistical risk scores that predict the risk of patients with previous CVD event or non-CVD event, including SCORE2 (Systematic COronary Risk Evaluation 2) [5], QRISK3 [6], Framingham Risk Score (FRS) [7], Joint British Society risk calculator 3 (JBS3) [8], Atherosclerotic Cardiovascular Disease (ASCVD) [9], American College of Cardiology/American Heart Association (ACC/AHA) [9], HeartScore [10], WHO risk score [11], and CoroPredict [12].

The effectiveness of the statistical models, risk scores, was evaluated in several studies [13-14] based on different ethnicities. The main outcome of these works was that these scores predict the percentage of patients to be at high or low risk of developing CVD.

Machine learning algorithm is an alternative effective approach, which can also be used for the detection of disease outcomes and events, when trained on proper medical data. Recently, the machine learning models were used extensively for the prediction of CVD [15-17]. The proposed machine learning models achieve good predictions of CVD, with accuracy larger than 90%. More specifically, in [18], the hyOPTXg model extracted the highest AUC value equal to 0.947, with the use of optimization techniques (min-max scaling, OPTUNA: Hyper-parameter tuning) and an optimized Extreme Gradient Booster. In another study [19], in which ten ML models were applied, the Extreme Gradient Boost and Gradient Boost presented the highest AUC value (0.812). Also, compared to Framingham and ACC/AHA risk models, the ML models presented equal to or greater outcomes.

In addition, Pouriyeh *et al*. [20], used different machine learning approaches including Naıve Bayes (NB), Multilayer Perceptron (MLP), Decision Tree (DT), K-Nearest Neighbor (K-NN), Radial Basis Function (RBF), Support Vector Machine (SVM) and Single Conjunctive Rule Learner (SCRL), on the Cleveland Heart Disease database. They applied the bagging, boosting and stacking, in order to evaluate the performance of each classifier and such classifiers in combination. The comparison of all classifiers shows that SVM achieved the highest accuracy and SCRL has the lowest

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accuracy with 84.15% and 69.96%, respectively. Furthermore, SVM is still the best approach with the same accuracy percentage, whenever the bagging method is applied. However, in this case, DT is the worst model with 78.54%. SVM has also improved to 84.81% with the use of bagging. The combination of classifiers, MLP and SVM, proved to be the most accurate with 84.15%, after the stacking.

In this work, we aim to predict death caused by CVD after ten years of follow-up using only simple biomarkers which are routinely collected in the Ludwigshafen Risk and Cardiovascular Health (LURIC) dataset [21]. In particular, we applied six different machine learning algorithms from which the best performance is presented by Logistic Regression. The results shows that the mean values of accuracy and area under receiver operative characteristic curve 72.20 % and 72.97 %, respectively.

II. MATERIALS AND METHODS

A. Overall workflow

Fig. 1 depicts the workflow of this analysis. We started with LURIC dataset, which includes patients scheduled for coronary angiography. This was followed by preprocessing the data with cleaning and feature selection techniques. Subsequently we utilized ML algorithms and Class Imbalance handling. As a result of that, we calculated the performance of ML algorithms in predicting death by CVD. These results are part of the TIMELY study that aims to estimate the risk score and mortality of CVD in patients with 10-year risk.

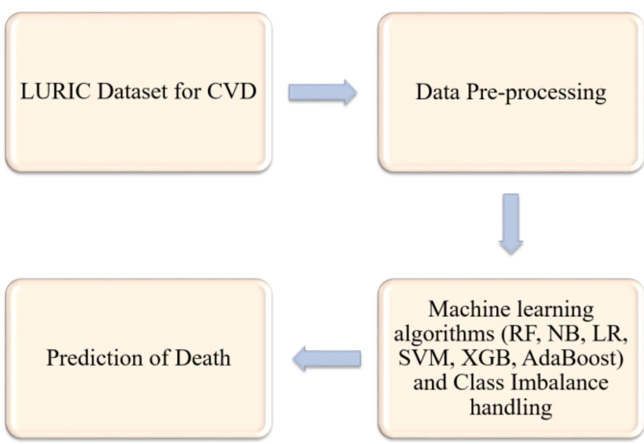

Figure 1. Working diagram of current study.

B. Data description

In this study, the medical data used were from the Ludwigshafen Risk and Cardiovascular Health (LURIC) cohort [21]. It contains 3,316 patients and 3,023 features including demographics, daily habits, biomarkers, such as inflammatory and molecular, genomics, T-cells, antibodies, and interleukins were recorded. It also includes annotation of death caused by CVD after 10 years. Thus, annotation is displayed as the target in our study and is classified into three classes. Class zero represents alive patients, class one is the dead patients due to CVD, and class three displays dead patients due to another disease. We took into account only the class zero and class one, with 2547 patients and 484 patients, respectively. Moreover, we included 20 features in the analysis which are collected, all associated with cardiovascular disease. All features were selected with a feature selection technique, which is based on the importance of features. These features and their mean values are presented in Table I. The clinical data used in this analysis are easily accessible to the clinician. This means that the presented results correspond to the daily clinical practice.

TABLE I. THE DETAILED DATASET ATTRIBUTES DESCRIPTION FOR DATASET (LURIC).

No.	Attributes			
	Features	***Type***	***Mean/ Percentage***	***St. Deviation***
1	Age (years)	Numeric	61.98	10.60
2	Sex	Binary	Male=69.2% Female=30.8%	-
3	Weight (kg)	Numeric	79.97	13.69
4	Total Cholesterol (mg/dL)	Numeric	192.63	39.18
5	HDL Cholesterol (mg/dL)	Numeric	38.84	10.88
6	LDL Cholesterol (mg/dL)	Numeric	116.85	34.42
7	Cholesterol (mg/dL)	Numeric	208.81	43.96
8	Triglycerides (mg/dL)	Numeric	172.03	115.30
9	LDL Triglycerides (mg/dL)	Numeric	31.28	11.73
10	HDL Triglycerides (mg/dL)	Numeric	15.82	7.03
11	Type II Diabetes Mellitus	Binary	No=83.4% Yes=16.6%	-
12	Urea (mg/dl)	Numeric	39.08	14.96
13	Uric Acid (mg/dl)	Numeric	5.09	1.68
14	Glycosylated hemoglobin (%)	Numeric	6.29	1.23
15	Interleukin-6 (ng/L)	Numeric	12.41	12.70
16	Oxidized LDL	Numeric	74.76	28.30
17	History of arterial hypertension	Binary	No=42.0% Yes=58.0%	-
18	Heart rate	Numeric	68.52	12.27
19	Systolic Blood Pressure	Numeric	140.47	24.60
20	Diastolic Blood Pressure	Numeric	80.94	12.31

C. Data pre-processing

The pre-processing of data has been one of the major steps in our analysis. The techniques applied include cleaning (SimpleImputer class), feature scaling, feature selection (SelectKBest method), and under sampling for making the balanced dataset.

In our study, we used the down sampling method. This method is applied when the sample size of data is sufficient [22]. We kept all samples in the minority class one and randomly selecting an equal number of samples in the majority class zero. This procedure is finished until the majority and minority class observations are balanced to produce a new balanced 1:1 ratio dataset for further analysis. Eventually, the analysis was conducted in 484 dead patients and 484 random alive patients. Further to this process, the dataset was divided into training and test set (40 %).

D. Supervised machine learning algorithms

In the current study, six supervised machine learning classifiers were chosen, including the Random Forest (RF) [23], Logistic Regression (LR) [24], Support Vector Machine (SVM) [25], Naïve Bayes (NB) [26], Extreme Gradient Boost (XGB) [27] and Adaptive Boost (AdaB) [28]. These models are widely accepted as fit for risk prediction.

E. Performance evaluation metrics

In our analysis, we estimated the performance using standard metrics, namely Accuracy (ACC), Precision, F1-Score, Sensitivity/Recall, and Specificity. Furthermore, the Receiver Operative Characteristic Curve (ROC) and the Area under the ROC curve (AUC) have been employed to compare the performance of each classifier.

III. Results

The results were extracted using the python software package for all classification models. Especially, we utilized the robust library scikit-learn 1.0.2, which provides efficient tools for predictive data analysis. All values of the measurements are shown and compared in Table II. The presented values are the mean values across ten runs.

TABLE II. Results Of The Pediction Performance For Each Model

Models	Performance Evaluation				
	ACC (%)	*Prec*(%)*	*Recall (%)*	*F1-Score (%)*	*Spec^(%)*
RF	63.10	76.1	68.4	72.6	76.1
SVM	70.70	73.4	66.8	70.3	73.9
NB	65.21	77.2	48.5	59.4	83.7
LR	72.20	77.5	68.7	72.9	77.0
XGB	72.06	75.4	69.7	72.3	74.3
AdaBoost	64.30	72.3	62.0	68.3	71.7

* Prec=Precision (%), ^Spec=Specificity

Furthermore, the mean values of accuracy are shown in Fig. 2. The results of experiments show that Logistic Regression and Random Forest have 72.20% and 63.10 % respectively, the highest and lowest accuracy of all the classifiers tested. The Logistic Regression classifier is the most suitable model to predict the mortality of CVD in patients with a 10-year history of CVD.

The area under receiver operating characteristic curve (AUC) has also been calculated for all applied machine learning algorithms are shown in the Fig. 3. A value of AUC close to 1 signifies that the model has the highest accuracy. Comparing the AUC measurements, we observe that the AUC value of Logistic Regression model is the highest, and the AUC value of Naïve Bayes model is the lowest.

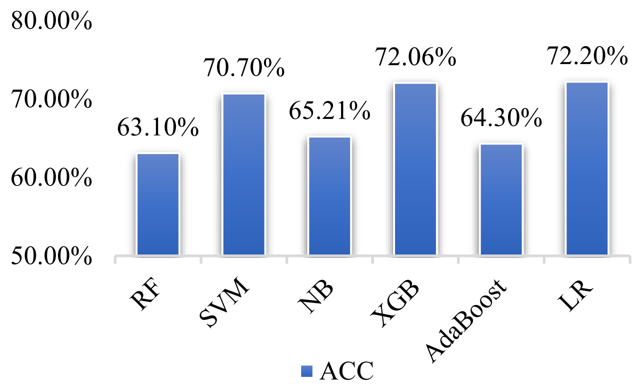

Figure 2. The mean accuracy values for six classifiers.

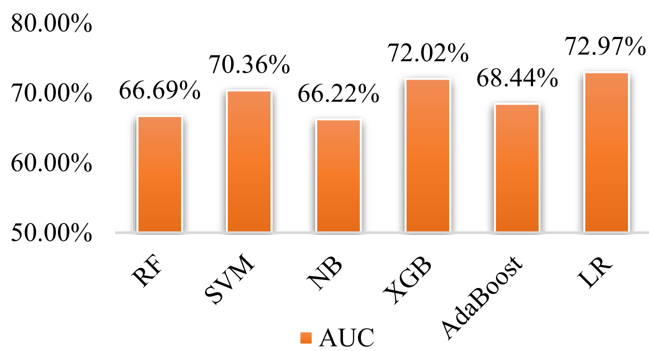

Figure 3. The mean values of area under receiver operative characteristic curve for six different classifiers.

In addition, the visualization of the results can be achieved using a ROC curve analysis. We plotted the ROC curves for each classifier in an indicative run of the algorithm (Figure 4). Furthermore, we estimated the receiver operating characteristic (ROC) curve to estimate the efficiency of the applied models. The indicative run of the algorithm shows that Logistic Regression and Support Vector Machine achieved approximately equal AUC values of 0.726 and 0.706, respectively.

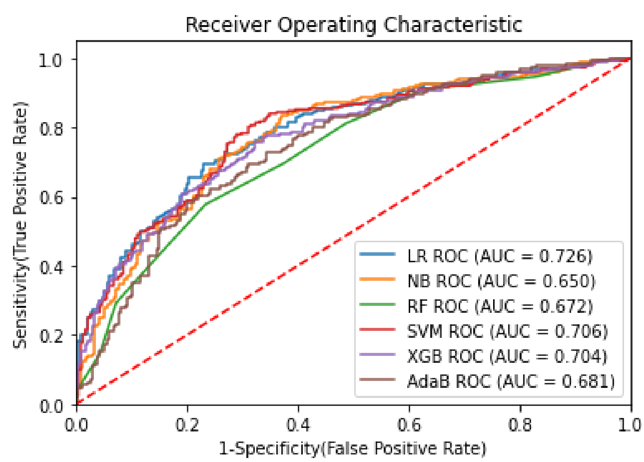

Figure 4. Receiver operating characteristic curve of the prediction performance for every classifier in an indicative algorithm run.

IV. Conclusion

Monitoring patients with a history of cardiovascular disease is the most important method for preventing new cases and reducing patient mortality. In this study, the 10-years risk of cardiovascular mortality of patients suffering scheduled for angiography has been predicted utilizing machine learning techniques. The six machine learning models have been applied and tested with various values of parameters to get the highest accuracy before the comparison has been made. The efficiency of each classifier has been evaluated. Among the six different techniques, Logistic Regression outperformed all the others, by having the highest average accuracy (72.20 %) and AUC value (72.97 %). In this paper, the novelty stands in the parallel use of multiple ML models, including both modern (e.g. XGB) and standard algorithmic models (e.g. LR), aiming to predict the 10-years CVD caused mortality using only easily collected biomarkers in the everyday clinical routine. Our principal goal has been to predict the mortality of CVD using different machine learning techniques in a small dataset and to choose the best predictive computational model comparing them.

There were limitations such as the population size and the absence of optimization techniques. This paper is based on 10 consecutive runs, resulting in LR mean accuracy and AUC values slightly surpassing those of XGB. We aim to include 100 runs in a future analysis to further increase the accuracy and validity of the results. Finally, a future goal is to establish a risk score. In addition, the established Coropredict score will be estimated and compared with its calculated values in the LURIC dataset within the TIMELY study.

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